

KRITANK SINGH

Lead Data Engineer · Citicorp Services India Pvt. Ltd.
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SUMMARY

Data Engineer with 5 years at Citi building and optimizing distributed data systems that process terabytes in production. Delivered \$1M in annual cost savings by redesigning ETL logic across 500+ extracts, cut a critical pipeline from 8+ hours to under 2, and led a zero-downtime Cloudera CDH to CDP platform migration. Deep Apache Spark internals expertise across Scala and PySpark: execution plan tuning, skew mitigation, join strategy selection, and shuffle optimization on multi-terabyte Hive and HDFS datasets.

SKILLS

Distributed Processing: Apache Spark (Scala & PySpark, Spark 3), Spark SQL, Spark Structured Streaming, Hive, HiveQL, YARN
Languages: Python, Scala, SQL, Shell/Bash, Java, C++ **Storage:** HDFS, Parquet, ORC, Avro, Delta Lake, AWS S3, Kafka (fundamentals)
Platforms & Tools: Cloudera CDH/CDP, Databricks, AWS, Apache Airflow, Autosys, Connect Direct, Jenkins, Git, Bitbucket, Jira
Data Engineering: ETL/ELT Pipeline Design, Performance Tuning, Lakehouse Architecture, Data Quality Frameworks, Delta Processing, Platform Migration

EXPERIENCE

- Lead Data Engineer

Pune, India

Citicorp Services India Private Limited (CSIPL) – Technology Division of Citigroup

Apr 2025 – Present

- Own the **risk-score pipeline chain** serving downstream risk models, processing **multi-terabyte** Hive/Parquet datasets on HDFS through skew-heavy join sequences tuned via broadcast thresholds, adaptive query execution, and salted partitioning.
- Led redesign of ETL logic across **500+ production extracts**, consolidating redundant scans and rewriting execution plans, delivering **\$1 million in annual cost savings**.
- Cut extract generation and transmission from **8+ hours to under 2 hours (75% reduction)** by re-partitioning on join keys, eliminating wide shuffle transformations, and right-sizing executor memory and shuffle partition counts.
- Spearheaded **platform migration from Cloudera CDH to CDP** with **zero downtime** across 500+ extracts, running parallel validation to certify output parity before cutover.
- Built a reusable **delta processing framework** driven by configurable column lists, replacing bespoke per-pipeline change detection logic across multiple teams.
- Driving the team's **Spark Scala to PySpark (Spark 3) migration**, authoring the shared standardization framework and coding conventions adopted across pipelines.

- Senior Data Engineer

Pune, India

Citicorp Services India Private Limited (CSIPL) – Technology Division of Citigroup

Jul 2022 – Mar 2025

- Built a **Spark-based extract validation framework** performing order-independent schema and row-level reconciliation, catching defects pre-production and shortening release cycles.
- Resolved **data skew** and optimized reads of **80GB+ Parquet part files** on HDFS through partition strategy, join-side tuning, and adaptive execution.
- Designed and maintained **secure data exchange pipelines** with external partners and vendors over Connect Direct, with automated retry, checksum verification, and audit logging.
- Ran **technology POCs** on Databricks and Delta Lake, driving adoption of the resulting patterns into production pipelines.

- Data Engineer

Pune, India

Citicorp Services India Private Limited (CSIPL) – Technology Division of Citigroup

Jul 2021 – Jul 2022

- Integrated ETL processes with a **centralized audit framework** across **1000+ server-based jobs**, creating a single source of truth for job-level lineage and reconciliation.
- Improved **Spark query and job performance** via partitioning, broadcast joins, and execution plan analysis, materially reducing runtime on critical-path jobs.

PROJECTS

- Creator Analytics Platform

Python, PySpark, AWS, Instagram Graph API, YouTube Data API v3

2025 – Present

- Built the ingestion layer for nightly batch pulls via the Instagram Business Discovery endpoint, using field batching, caching, and token lifecycle management to stay within API rate limits.
- Designed the warehouse schema (posts, comments, feature store, profile snapshots, benchmarks) and a **PySpark gap-scoring engine** computing per-creator performance deltas against tier-and-category benchmarks to produce ranked recommendations.
- Dynamic YARN queue selection** (Bash): job-submission wrapper that inspects live queue capacity and routes Spark jobs to the least-contended queue, cutting scheduling wait on a shared on-premises Hadoop cluster.

AWARDS

- Citi Gold Award**– Led the CDH-to-CDP platform migration, delivered with zero downtime across 500+ extracts.
- Citi Silver Award**– Exemplary performance across multiple high-priority parallel projects.
- Citi Bronze Award**– Consistent delivery and contribution to team objectives.

TRAINING

- Apache Spark 3** – Spark SQL and PySpark 3, Spark Programming in Scala, Beyond Basics (Udemy)
- Platform & Tooling** – Databricks Fundamentals and Apache Spark Core, Apache Airflow, AWS DVA-C02 exam preparation (Udemy)

EDUCATION

- Motilal Nehru National Institute of Technology (MNNIT), Allahabad

Bachelor of Technology, Computer Science and Engineering | CGPA: 8.8 / 10

Prayagraj, India

2017 – 2021